

Probability Basics for Machine Learning

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1 Probability

Definition 1.1 (Probability space). A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where

- Ω is the *sample space*;
- $\mathcal{F} \subseteq 2^\Omega$ is a σ -algebra (contains Ω and is closed under complementation and countable unions);
- $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a probability measure satisfying the Kolmogorov axioms:
 1. *Nonnegativity*: $\mathbb{P}(A) \geq 0$ for all $A \in \mathcal{F}$;
 2. *Normalization*: $\mathbb{P}(\Omega) = 1$;
 3. *Countable additivity*: for pairwise disjoint $A_i \in \mathcal{F}$, we have $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

Basic properties.

- Monotonicity: $A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$;
- Complement: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$;
- Inclusion-exclusion: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$;
- Subadditivity: $\mathbb{P}(\bigcup_i A_i) \leq \sum_i \mathbb{P}(A_i)$;
- Continuity: if $A_n \uparrow A$ then $\mathbb{P}(A_n) \uparrow \mathbb{P}(A)$, and if $A_n \downarrow A$ then $\mathbb{P}(A_n) \downarrow \mathbb{P}(A)$.

Conditional probability, total probability, and Bayes' rule.

- Conditional probability: For $\mathbb{P}(B) > 0$, $\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$, which also implies that $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B)$.
- Total probability: If $\{B_i\}$ is a (countable) partition of Ω with $\mathbb{P}(B_i) > 0$, then $\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$.
- Bayes' rule: For any partition $\{B_i\}$, $\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}$.

Independence. Events A, B are independent if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

Random Variable.

Definition 1.2 (Random Variable). A *random variable* is a function $X : \Omega \rightarrow \mathbb{R}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ that maps outcomes in the sample space Ω to real values.

Definition 1.3 (Distribution). The *distribution* of a random variable X is specified by its cumulative distribution function (CDF), which is given by $F_X(x) = \mathbb{P}(X \leq x)$, for a continuous random variable. If X is discrete, the distribution is given by the probability mass function (PMF), $p_X(x) = \mathbb{P}(X = x)$.

Definition 1.4 (Expectation). The *expectation* (or mean) of a random variable X is the integral (or sum for discrete variables) of X with respect to its probability measure. It is defined as

$$\begin{aligned}\mathbb{E}[X] &= \int_{\Omega} X(\omega) d\mathbb{P}(\omega), \quad \text{for continuous variables,} \\ \mathbb{E}[X] &= \sum_{\omega \in \Omega} X(\omega) \mathbb{P}(\omega), \quad \text{for discrete variables.}\end{aligned}$$

Definition 1.5 (Variance). The *variance* of a random variable X measures the spread of its values around the mean. It is defined as

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Alternatively, the variance can be expressed as

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Definition 1.6 (Covariance). The *covariance* between two random variables X and Y measures the degree to which they vary together. It is defined as

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])],$$

or equivalently,

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Properties of Expectation.

- **Linearity:** The expectation operator is linear, meaning that for any random variables X and Y , and constants a and b ,

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

- **Non-negativity:** If $X \geq 0$ almost surely, then

$$\mathbb{E}[X] \geq 0.$$

- **Expectation of a constant:** If c is a constant, then

$$\mathbb{E}[c] = c.$$

- **Expectation of a random variable:** If g is a measurable function and X is a random variable, then

$$\mathbb{E}[g(X)] = \int_{\Omega} g(X(\omega)) d\mathbb{P}(\omega)$$

for random variables. Specifically, we have

$$\mathbb{E}[g(X)] = \sum_{\omega \in \Omega} g(X(\omega)) \mathbb{P}(\omega)$$

for discrete random variables.

Properties of Independent Random Variables.

- **Independence of Random Variables:** Two random variables X and Y are independent if for all sets A and B ,

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A) \mathbb{P}(Y \in B).$$

This means that the occurrence of one event does not affect the probability of the other.

- **Sum of Independent Random Variables:** If X and Y are independent random variables, then the variance of the sum is the sum of the variances:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

- **Product of Independent Random Variables:** If X and Y are independent random variables, then the expectation of their product is the product of their expectations:

$$\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y].$$

- **Independence and Covariance:** If X and Y are independent random variables, then their covariance is zero:

$$\text{Cov}(X, Y) = 0.$$

- **Independence and Joint Distribution:** If X and Y are independent random variables, then their joint probability distribution factors as the product of their marginal distributions:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y),$$

where $f_{X,Y}(x, y)$ is the joint probability density function (for continuous random variables), and $f_X(x)$, $f_Y(y)$ are the marginal probability density functions.

Conditional Expectation.

- **Linearity of Conditional Expectation:** For random variables X and Y , and constants a and b ,

$$\mathbb{E}[aX + bY \mid Z] = a\mathbb{E}[X \mid Z] + b\mathbb{E}[Y \mid Z].$$

- **Taking out measurable variables:** If Z is measurable with respect to a σ -algebra $\mathcal{G}$, then

$$\mathbb{E}[ZX \mid \mathcal{G}] = Z\mathbb{E}[X \mid \mathcal{G}].$$

- **Iterated Expectations (Law of Total Expectation):** If Y is a random variable, then

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]].$$

This law states that the expectation of Y can be computed by first conditioning on another random variable X , and then taking the expectation with respect to X .

- **Conditional Expectation and Independence:** If X and Y are independent, then

$$\mathbb{E}[X \mid Y] = \mathbb{E}[X].$$

This result is intuitive, as conditioning on Y does not provide any additional information about X when they are independent.